Lab 8:- Encoder and Decoder Implementation

Objectives:

- To understand the working principle of encoder and decoder circuits and how they convert information between diff. binary formats.
- To design and implement encoder and decoder circuits using Logic gates and verify their operation using truth tables.

Theory:

Decoder

↳ A decoder is a logic circuit that accepts a sets of inputs that represents a binary number and activates only the output that corresponds to the input number.

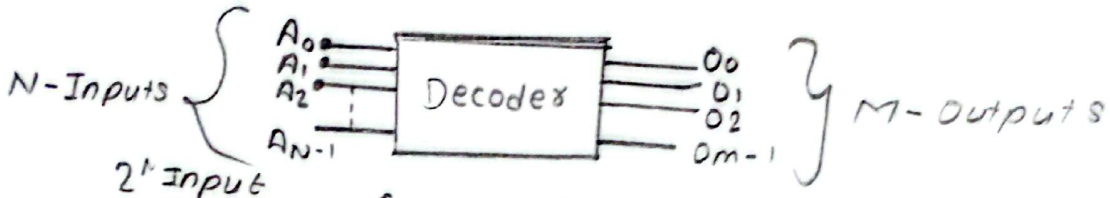

fig:- Block diagram of Decoder

3-8 Line decoder

3 to 8 line decoder (active high) is a combinational circuit that converts 3 input lines into 8 output lines:

Truth Table:-

A	B	C	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

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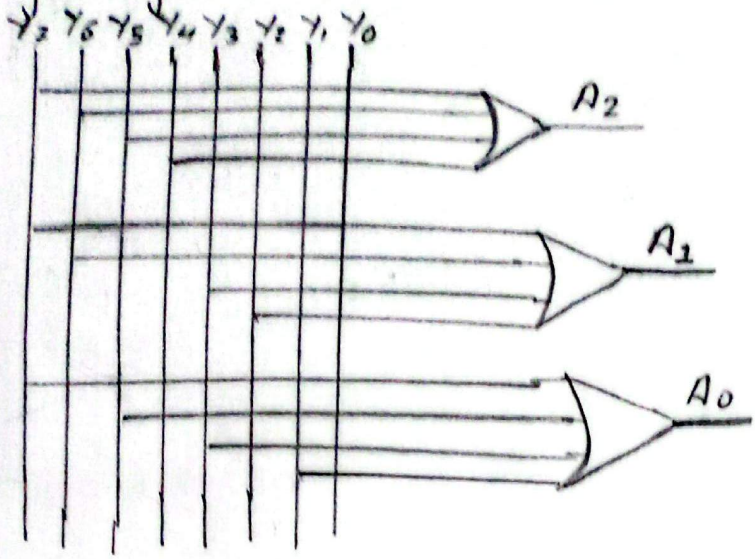
Logic diagram :-



Observation.

1.) 3-8 line decoder (Active High)

Input			Observed output								Expected Output								
A	B	C	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_1	D_0
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0
0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Here, expected output and observed output are same.

The corresponding Boolean expression are.

$D_0 = A'B'C'$

$D_1 = A'B'C$

$D_2 = A'BC'$

$D_3 = A'BC$

$D_4 = AB'C'$

$D_5 = AB'C$

$D_6 = ABC'$

$D_7 = ABC$

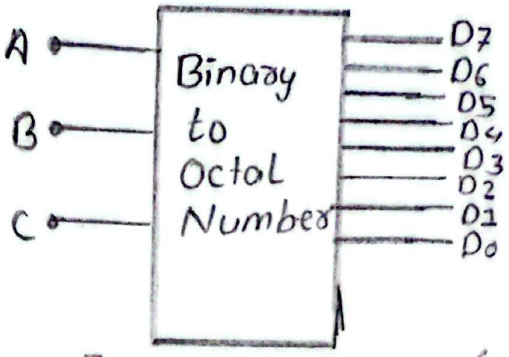
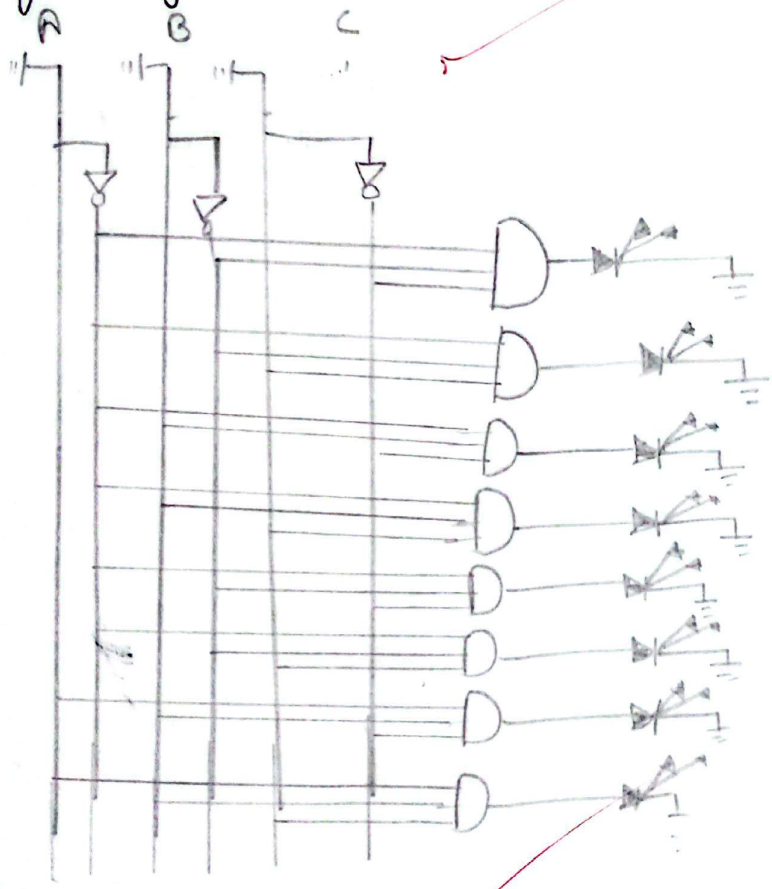


Fig :- Block diagram of 8-3 line Decorder

Logic diagram



Encoder:

An encoder is a combination circuit that ~~espential~~ essentially performs a 'reverse' of decoder functions. It has maximum of 2^n inputs ~~tik~~ lines and 'n' output lines. An encoder accepts an active level one of its inputs, representing digits such as decimal or octal digits and converts it to a coded outputs such as BCD or binary.

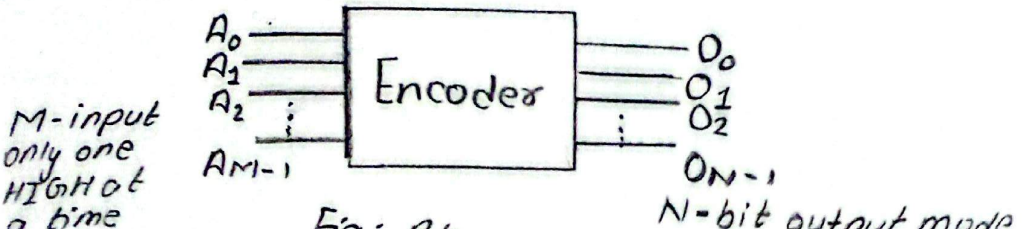


Fig: Block diagram of Encoder.

8-3 line Encoder

An 8 to 3 line encoder is a combinational circuit that converts 8 input lines into 3 output lines. It performs the reverse operation of decoder.

Truth Table:

Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

The Boolean Function for each output

$A_2 = Y_7 + Y_6 + Y_5 + Y_4$

$A_1 = Y_7 + Y_6 + Y_5 + Y_2$

$A_0 = Y_7 + Y_5 + Y_3 + Y_1$

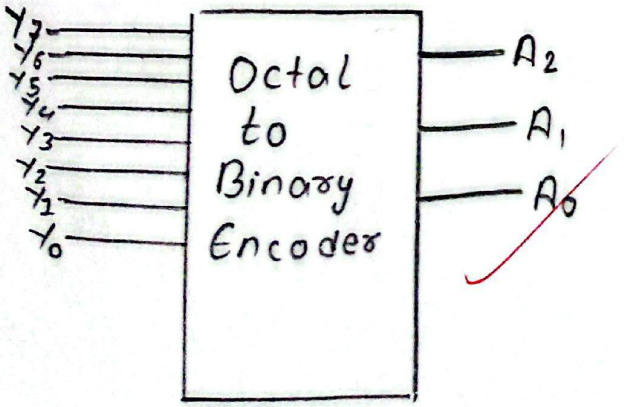


Fig: Block diagram of 8-3 Encoder.

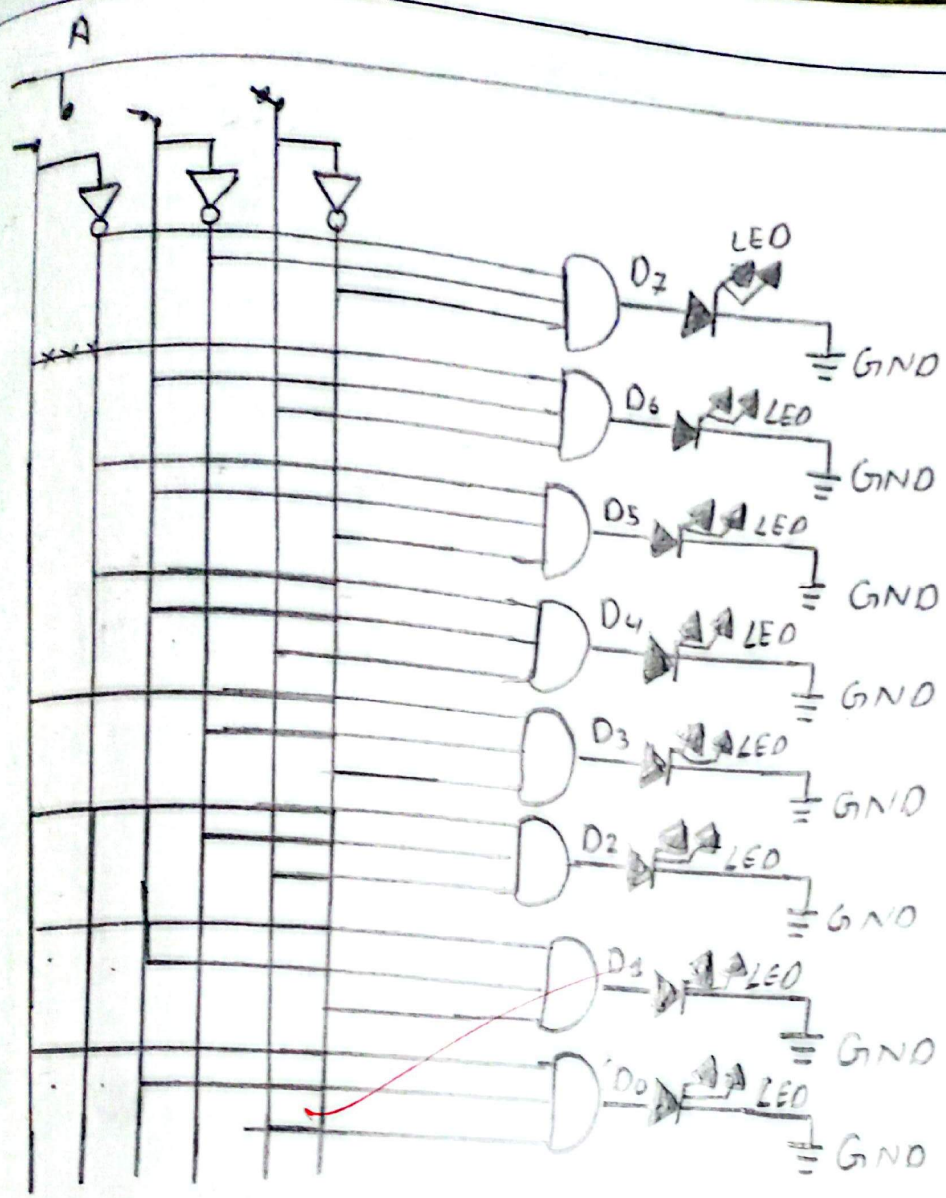


Fig :- Combination Circuit of decoder

2.) 8 to 3 Line Encoder (Active High)

Input								Expected			Observed		
Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	1	0	0	1
0	0	0	0	0	1	0	0	0	1	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1	0	1	1
0	0	0	1	0	0	0	0	1	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1	1	0	1
0	1	0	0	0	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1	1	1	1

Here, the expected output and observed output are same.

Conclusion

In this lab session, we learned about decoder and encoder and there theorys.

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